

Problem 1.9

Using compound interest formula, what annual interest rate would cause an investment of \$5,000 to increase to \$7,000 in 5 years?

Solution

Using the compound interest formula

$$A = P(1 + i)^t,$$

substitute the given values:

$$7000 = 5000(1 + i)^5.$$

Divide both sides by 5000:

$$\frac{7000}{5000} = (1 + i)^5.$$

So,

$$1.4 = (1 + i)^5.$$

Taking the fifth root of both sides,

$$1 + i = (1.4)^{1/5}.$$

Therefore,

$$i = (1.4)^{1/5} - 1.$$

Calculating,

$$i \approx 0.06961.$$

Hence, the annual interest rate is approximately

$$\boxed{6.961\%}.$$